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Editors-in-Chief
Smart Agricultural Technology

Dear Editors,

Please consider the manuscript, "Support-Aware Review Queues for Crop–Weed Vision: A Multitemporal Single-Field Case Study," for publication in *Smart Agricultural Technology*.

The study addresses a practical commissioning question for agricultural vision: which stage currently limits a crop–weed review queue, candidate generation, role qualification, rank ordering, or allocation of a finite review budget? We answer this question on the official WeedsGalore spatial split using instance-level denominators from proposal formation through final review selection.

The framework separates spatial proposal recall, role-qualified recall, and queued recall under many-to-one and one-to-one matching. Shared-ranker-weight comparisons isolate proposal formation, while all-candidate fitting aligns ranker supervision with queue support. Candidate-level role-qualified AUC and average precision are linked to instance recovery, crop exposure, and the finite review budget rather than treated as end-to-end system metrics.

A target-semantic baseline with equal target-mask access further distinguishes component formation from ranking. Across five independent H200 training seeds, fixed-K precision and one-to-one recall were 0.7168 ± 0.0260 and 0.2027 ± 0.0057. Exploratory 5–50 candidates/tile allocation improved both metrics in every seed, to 0.8939 ± 0.0249 and 0.2578 ± 0.0090, while worst-date recall decreased in every seed. A complete 16-policy quota grid and date-balanced comparator show how pooled utility, weakest-date coverage, and spatial-block effects diverge. Validation selection retained fixed $K = 20$ in four seeds and 0–30 allocation in one.

The manuscript presents an application-focused queue evaluation as a multitemporal single-field case study. Its chronology distinguishes the archived core analysis from post-test exploratory extensions. The accompanying package includes editable figures, source-data tables, exact component-mask matching outputs, complete proposal and quota grids, versioned configurations, failure records, run hashes, runtime and memory records, and clean-room replay commands.

Sincerely,

Zhuo Chen
on behalf of all authors